

Supplementary Material for

“Surplus Allocation in Energy Community Games under Non-convex Operations: A Duality-based Perspective”

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This supplementary material has two parts. Section S1 collects the data of the numerical study, that is, the parameters of the community dispatch model, the asset composition of the four communities, and the asset matrix of each member. Section S2 writes out the two-stage stochastic extension. Section, equation and proposition numbers without an “S” prefix refer to the main paper.

S1. DATA OF THE NUMERICAL STUDY

This section reports the data of the numerical study in three parts: the parameters of the dispatch model, which are common to all four communities; the asset composition of each community; and the asset matrix of each individual member.

A. Parameters of the Community Dispatch Model

Table S1 reports every parameter of the dispatch model of Section III-B of the main paper. All entries are held fixed across the four community sizes, so that the size axis varies n alone. The electrolyzer is a three-state unit. Besides on and off it holds a standby state that draws 1% of its rating, which is cheaper than a cold restart but not free, and a return from off costs 50 €. Its minimum load is 15% of capacity, which lies within the range of 10 to 16% reported for PEM units. The standby draw is below the 2.5 to 5% reported in the same literature, so the unit is modelled as cheaper to park than a typical installation. If anything this understates the value of commitment and with it ω^{LR} . The power-to-hydrogen curve is concave and is represented exactly by six segments with segment-selection binaries. The heat pump is a two-state unit. It is committed between 20% and 80% of its rating at a constant coefficient of performance of 3.28, and each start costs 10 €. Its reserve headroom is available only while it is on, which is why its contribution is measured in electric MW while its operating window is on heat output. Storage is sized relative to its owner rather than in absolute terms. Rated power is a quarter of the asset capacity of the owner and energy capacity is three times that power, so the storage of a member scales with the assets it operates. Demand is that of 700 households. The community is a price taker and faces an import price 1.5 times its export price.

B. Composition of the Four Communities

Table S2 counts the members that own each asset type at each community size. The $n = 60$ community doubles every

TABLE S1
PARAMETERS OF THE COMMUNITY DISPATCH MODEL. HELD FIXED
ACROSS EVERY COMMUNITY SIZE, SO THAT THE SIZE AXIS VARIES n
ALONE.

Parameter	Value	Note
<i>Market and horizon</i>		
Horizon $ T $	24 h	hourly resolution
Households	700	
Import/export price ratio	1.5	$\pi^{\text{imkt}} / \pi^{\text{emkt}}$
H ₂ reference price	3.33 €/kg	
Price data	Jeju (KR) and DK2, January	
<i>Electrolyzer (3-state commitment)</i>		
Rated draw $\bar{d}^G$	1.0 MW	
Minimum draw $\underline{d}^G$	0.15 MW	0.15 × rated
Standby draw c_j^{sb}	0.01 MW	
Start-up cost $c_j^{G,su}$	50 €	
Minimum down time	2 h	
Efficiency curve	piecewise linear, 6 segments	
<i>Heat pump</i>		
Capacity $\bar{c}^H$	0.8 MW _{th}	
Coeff. of performance ν^{COP}	3.28	el. to heat
<i>Storage (per carrier k)</i>		
Power rating	0.25	× owner's capacity
Energy capacity	3.0	× power rating
Initial state of charge	0.2	of energy capacity
Round-trip eff., E / H ₂	0.95	each way
Round-trip eff., heat	0.90	each way
<i>Reserve and peak coupling</i>		
Reserve price π^{res}	56 €/MW·h	FCR-N, DK2
Reserve product	symmetric, $ I = 6$ blocks of 4 h	
Peak penalty δ^{peak}	150 €/MW	

count at $n = 30$ exactly, so these two sizes differ in n and in nothing else. The smaller communities do not match them in every share, because the asset mix at $n = 6$ and $n = 15$ was set by role rather than by proportion. What is held across all four sizes is the last row, which counts the members that carry commitment binaries and stays near one third. These are the members that generate ω^{LR} , so the flatness of ω^{LR} in n is not confounded by a non-convex share that drifts with n .

C. Member Configurations

Table S3 reports the asset matrix of every member. The four communities are nested, so the first 6, 15 and 30 rows are the $n = 6$, $n = 15$ and $n = 30$ communities unchanged. At $n = 6$ every function is held by a single archetype. Member u1 owns wind with electric storage, u2 an electrolyzer, u3 a heat pump, and u4 to u6 are non-flexible consumers, one per carrier. Every

TABLE S2

COMPOSITION OF THE FOUR COMMUNITIES. ENTRIES ARE THE NUMBER OF MEMBERS OWNING THE ASSET, WITH THE SHARE OF THE COMMUNITY IN BRACKETS, IN PERCENT. THE LAST ROW COUNTS THE MEMBERS CARRYING COMMITMENT BINARIES, WHICH ARE WHAT GENERATES ω^{LR} .

	$n = 6$	$n = 15$	$n = 30$	$n = 60$
Wind	1 (16.7)	2 (13.3)	3 (10.0)	6 (10.0)
Solar	–	3 (20.0)	8 (26.7)	16 (26.7)
ELZ	1 (16.7)	3 (20.0)	6 (20.0)	12 (20.0)
HP	1 (16.7)	3 (20.0)	4 (13.3)	8 (13.3)
E-sto	1 (16.7)	4 (26.7)	7 (23.3)	14 (23.3)
H ₂ -sto	1 (16.7)	2 (13.3)	3 (10.0)	6 (10.0)
Q-sto	1 (16.7)	1 (6.7)	2 (6.7)	4 (6.7)
D^E	1 (16.7)	4 (26.7)	10 (33.3)	20 (33.3)
D^{H_2}	1 (16.7)	2 (13.3)	6 (20.0)	12 (20.0)
D^Q	1 (16.7)	3 (20.0)	7 (23.3)	14 (23.3)
Commitment binaries	2 (33.3)	6 (40.0)	10 (33.3)	20 (33.3)

coalition value at that size can therefore be traced to a named role. The members added at $n = 15$ and $n = 30$ introduce competition within each role, members that own generation and conversion together, standalone storage operators, and consumers active in two carriers. The last column of Table S3 gives the capacity multiplier applied to each member added at $n = 60$. Every added member differs from the archetype it mirrors, so the community is not a two-fold replica of the 30-prosumer one. A replica would shrink the core toward the competitive allocation by the Debreu–Scarf theorem, and the comparison across sizes in the main paper would measure that effect instead of the effect of community size.

S2. TWO-STAGE STOCHASTIC EXTENSION

Remark 1 of the main paper states that relaxing Assumption 1(iv) leaves the construction of Section IV intact. This section writes out the dispatch problem and the Dantzig–Wolfe reformulation to which that remark refers, and verifies the claim.

A. Decision Stages

Let Ω be a finite set of scenarios of the renewable production, the load, and the external market prices, with probabilities $\rho_\omega > 0$ and $\sum_{\omega \in \Omega} \rho_\omega = 1$. Every parameter the scenario carries is indexed accordingly: the production and load bounds of (11g) are read as $\underline{p}_{jt}^{k,\omega}$, $\bar{p}_{jt}^{k,\omega}$ and $\underline{d}_{jt}^{k,load,\omega}$, $\bar{d}_{jt}^{k,load,\omega}$, the first of these carrying the renewable availability and the second the load. The electrolyzer states $z_{jt}^{G,on}$, $z_{jt}^{G,sh}$, $z_{jt}^{G,off}$ with their transition indicators, and the community reserve capacities r_i^{sym} , are decided before the realization. Every remaining decision is taken after it: the electrolyzer is redispached within its committed state, so that the segment selectors $\hat{z}_{jlt}^\omega$ and the segment inputs $\hat{d}_{jlt}^\omega$ carry a scenario index, and so do the heat pump input, the storage schedule, the reserve contributions, the community exchanges, and the peak p^ω .

Prices are revealed with the rest of the scenario. Import and export prices therefore carry a scenario index, $\pi_t^{k,imkt,\omega}$ and $\pi_t^{k,emkt,\omega}$, and so does the willingness to pay $\bar{u}_{jt}^{k,\omega}$, which in Section VI-A is the avoided cost of buying the carrier outside and so moves with the import price. Production costs $\bar{c}_{jt}^k$ and

the peak tariff δ^{peak} are technology and network parameters and stay deterministic. Assumption 1(iii) is read scenario by scenario, $\pi_t^{k,imkt,\omega} \geq \pi_t^{k,emkt,\omega}$ for every ω , which is what (13) needs in order to keep $\mathcal{X}_j^{MIP}$ bounded while leaving every coalition value unchanged. We further take the scenario prices to lie in a compact set fixed in advance, so that refining the scenario set adds outcomes without widening the range of prices any single outcome may take; Proposition S1(c) is stated under this reading.

The reserve capacity price is a special case. Since r_i^{sym} is sold before the realization and cannot be revised, a scenario-dependent $\pi_i^{res,\omega}$ enters the objective only through its mean $\bar{\pi}_i^{res} := \sum_{\omega \in \Omega} \rho_\omega \pi_i^{res,\omega}$, and its dispersion is immaterial to the model. We write $\bar{\pi}_i^{res}$ from here on.

Market trades are treated as recourse, which is the price-taking counterpart of Assumption 1(ii): the community adjusts its external position after the realization at the prices it faces. Its exposure to price uncertainty is therefore carried by the two first-stage decisions alone, the unit being committed and the reserve capacity sold before any price is known. Within a scenario the second stage sees the whole day at once, so the trading value it reports is an upper bound on what a multistage model would return. A day-ahead position settled against a separate balancing price is the sharper alternative. It adds a first-stage trade block to each prosumer and an imbalance term to the objective, and leaves every structural statement below unchanged, so we keep the simpler form.

B. Dispatch Problem

For a coalition $S \subseteq N$,

$$\begin{aligned}
 (DP_S^\Omega) : \max \sum_{\omega \in \Omega} \rho_\omega & \left[\sum_{j \in S} \sum_{k \in K} \sum_{t \in T} \left(\pi_t^{k,emkt,\omega} e_{jt}^{k,emkt,\omega} \right. \right. \\
 & - \pi_t^{k,imkt,\omega} i_{jt}^{k,imkt,\omega} + \bar{u}_{jt}^{k,\omega} d_{jt}^{k,load,\omega} - \bar{c}_{jt}^k p_{jt}^{k,\omega} \left. \right) - \delta^{peak} p^\omega \left. \right] \\
 & - \sum_{j \in S} \sum_{t \in T} c_j^{G,su} z_{jt}^{G,su} + \sum_{i \in I} \bar{\pi}_i^{res} r_i^{sym} \quad (S1a)
 \end{aligned}$$

subject to the community-level linking constraints, imposed in every scenario,

$$\sum_{j \in S} (i_{jt}^{k,com,\omega} - e_{jt}^{k,com,\omega}) = 0, \quad \forall t, k, \omega \quad (S1b)$$

$$r_i^{sym} \leq \sum_{j \in S} r_{jt}^{u,\omega}, \quad r_i^{sym} \leq \sum_{j \in S} r_{jt}^{d,\omega}, \quad \forall i, t \in T_i, \omega \quad (S1c)$$

$$\sum_{j \in S} (i_{jt}^{E,mkt,\omega} - e_{jt}^{E,mkt,\omega}) \leq p^\omega, \quad \forall t, \omega \quad (S1d)$$

and, for every $j \in S$, to the prosumer-level constraints (11e)–(11z) of the main paper, written for each ω with the first-stage indicators shared across scenarios and the bounds of (11g) taken at their scenario values. Four of them couple the two stages,

$$d_{jt}^{G,sc,\omega} = \tilde{d}_{jt}^{G,sc,\omega} + c_{jt}^{sh} z_{jt}^{G,sh}, \quad (S1e)$$

$$\underline{D}_{jt}^G \leq d_{jt}^{G,sc,\omega} \leq \bar{D}_{jt}^G, \quad (S1f)$$

$$\underline{d}_{jt}^G z_{jt}^{G,on} \leq \tilde{d}_{jt}^{G,sc,\omega} - r_{jt}^{u,els,\omega}, \quad (S1g)$$

$$\tilde{d}_{jt}^{G,sc,\omega} + r_{jt}^{d,els,\omega} \leq \bar{d}_{jt}^G z_{jt}^{G,on}, \quad (S1h)$$

$$z_{jt}^{G,on} = \sum_{l \in L} \hat{z}_{jlt}^\omega, \quad \tilde{d}_{jt}^{G,sc,\omega} = \sum_{l \in L} \hat{d}_{jlt}^\omega,$$

TABLE S3

MEMBER CONFIGURATION OF THE FOUR COMMUNITIES. LEFT PANEL, U1–U30; RIGHT PANEL, THE MEMBERS ADDED AT $n = 60$, WITH THE CAPACITY MULTIPLIER APPLIED TO EACH (APPLY_60PLAYER_OVERRIDES); A BLANK MULTIPLIER MEANS 1.0. HORIZONTAL RULES IN THE LEFT PANEL MARK THE $n = 6$ AND $n = 15$ COMMUNITIES. COLUMNS: W WIND, S SOLAR, EZ ELECTROLYZER, HP HEAT PUMP, $B_E/B_H/B_Q$ STORAGE AND $D_E/D_H/D_Q$ DEMAND IN ELECTRICITY, HYDROGEN AND HEAT.

Member	W	S	EZ	HP	B_E	B_H	B_Q	D_E	D_H	D_Q
u1	•				•					
u2			•							
u3				•						
u4								•		
u5									•	
u6										•
u7		•			•					
u8			•			•				
u9				•				•		
u10	•	•	•		•	•				
u11		•			•			•		
u12		•						•		
u13									•	
u14				•						•
u15								•		•
u16	•									
u17		•								
u18			•			•				
u19			•							
u20				•				•		
u21		•	•							
u22		•			•			•		
u23		•						•		•
u24					•			•		•
u25								•		•
u26		•							•	
u27									•	
u28								•	•	
u29					•			•	•	
u30										•

Member	W	S	EZ	HP	B_E	B_H	B_Q	D_E	D_H	D_Q	Cap. mult.
u31	•				•						R1.2, B_E 0.8
u32			•								EZ1.4
u33				•							HP0.7
u34								•			
u35									•		
u36										•	
u37		•			•						R0.7, B_E 1.3
u38			•			•					EZ0.9, B_H 1.4
u39				•				•			HP1.3, B_Q 0.8
u40	•		•		•	•					R0.7, EZ1.1, B_E 1.2, B_H 0.8
u41		•			•			•			R1.4, B_E 0.6
u42		•						•			R0.5
u43									•		
u44				•						•	HP1.2
u45								•		•	
u46	•										R1.4
u47		•									R0.6
u48			•			•					EZ0.7, B_H 0.9
u49			•								EZ1.3
u50				•				•			HP0.9, B_Q 1.2
u51		•	•								R1.1, EZ0.5
u52		•			•			•			R0.9, B_E 1.4
u53		•						•		•	R1.2
u54					•			•		•	B_E 0.7
u55								•		•	
u56		•							•		R0.8
u57									•		
u58								•	•		
u59					•			•	•		B_E 1.1
u60										•	

where $\underline{D}_{jt}^G, \overline{D}_{jt}^G$ are the window (12), which depends on the first-stage indicators alone. The last of these is where non-anticipativity is visible: the unit is on or off in every scenario alike, while the operating segment is chosen after the realization.

C. Compact Form

Stack the second-stage variables of prosumer j as $x_j = (x_j^\omega)_{\omega \in \Omega}$ and its binaries as $y_j = (z_j, (\hat{z}_j^\omega)_{\omega \in \Omega})$. The prosumer-level constraints are then

$$\mathcal{X}_j^{\text{MIP}} = \{(x_j, y_j) : T_j x_j + W_j y_j \leq h_j\}, \quad (\text{S2})$$

which is (14) of the main paper with T_j and W_j block-diagonal over Ω and with the rows (S1e)–(S1h) tying z_j to every block. It is bounded, since (11h) bounds each scenario block, and mixed-integer linear representable. The objective terms of prosumer j ,

$$f_j(x_j, y_j) = \sum_{\omega \in \Omega} \rho_\omega f_j^\omega(x_j^\omega, \hat{z}_j^\omega) - \sum_{t \in T} c_j^{G, su} z_{jt}^{G, su}, \quad (\text{S3})$$

are linear. Within a scenario each price multiplies a variable carrying the same scenario index, so no product of two variables appears; the objective would become bilinear only if the community set the prices, which Assumption 1(ii) excludes. The community-level variables and costs are

$$\begin{aligned} x_0 &= ((r_i^{sym})_{i \in I}, (p^\omega)_{\omega \in \Omega}) \geq 0, \\ c_0 &= ((\bar{\pi}_i^{res})_i, (-\rho_\omega \delta^{peak})_\omega), \end{aligned} \quad (\text{S4})$$

and x_0 appears on the left-hand side of (S1b)–(S1d) only. Stacking those rows, each equality as a pair of inequalities, gives $\sum_{j \in S} A_j x_j + A_0 x_0 \leq \sum_{j \in S} b_j$ with $b_j = 0$ and

$$m = \underbrace{2|K||T||\Omega|}_{(\text{S1b})} + \underbrace{2|T||\Omega|}_{(\text{S1c})} + \underbrace{|T||\Omega|}_{(\text{S1d})} = (2|K| + 3)|T||\Omega|. \quad (\text{S5})$$

Both sides are additive over the members, so Assumption 2 holds. No price appears in (S1b)–(S1d). The linking rows, and with them A_j , A_0 and the count m , are the same whether prices are known or not.

D. Relaxation and Reformulation

Dualizing (S1b)–(S1d) with $\theta \geq 0$, the contribution of x_0 is finite only on $\Theta = \{\theta \geq 0 : A_0^\top \theta \geq c_0\}$, which reads

$$\begin{aligned} \sum_{\omega \in \Omega} \sum_{t \in T_i} (\mu_t^{u, \omega} + \mu_t^{d, \omega}) &\geq \bar{\pi}_i^{res} \quad \forall i, \\ \sum_{t \in T} \xi_t^\omega &\leq \rho_\omega \delta^{peak} \quad \forall \omega. \end{aligned} \quad (\text{S6})$$

The relaxation then separates as in (19), with

$$v_j^{\text{LR}}(\theta) = \max_{(x_j, y_j) \in \mathcal{X}_j^{\text{MIP}}} \{f_j(x_j, y_j) - \theta^\top A_j x_j\}, \quad (\text{S7})$$

a two-stage stochastic mixed-integer program of a single prosumer. Let Q_j index the extreme points of $\text{conv}(\mathcal{X}_j^{\text{MIP}})$. A column $q \in Q_j$ is a complete contingent plan $(z_j^q, (x_j^{\omega, q}, \hat{z}_j^{\omega, q})_\omega)$,

with expected profit $w_j^q = f_j(\cdot^q)$ and linking contribution $a_j^q = A_j x_j^q \in \mathbb{R}^m$. The relaxed master is

$$(\text{DWR}_S^\Omega) : \max \sum_{j \in S} \sum_{q \in Q_j} w_j^q \lambda_j^q + \sum_{i \in I} \bar{\pi}_i^{\text{res}} r_i^{\text{sym}} - \sum_{\omega \in \Omega} \rho_\omega \delta^{\text{peak}} p^\omega \quad (\text{S8a})$$

$$\text{s.t. } \sum_{j \in S} \sum_{q \in Q_j} (i_{jt}^{k, \text{com}, \omega, q} - e_{jt}^{k, \text{com}, \omega, q}) \lambda_j^q = 0, \quad \forall t, k, \omega \quad (\text{S8b})$$

$$r_i^{\text{sym}} \leq \sum_{j \in S} \sum_{q \in Q_j} r_{jt}^{u/d, \omega, q} \lambda_j^q, \quad \forall i, t \in T_i, \omega \quad (\text{S8c})$$

$$\sum_{j \in S} \sum_{q \in Q_j} (i_{jt}^{E, \text{mkt}, \omega, q} - e_{jt}^{E, \text{mkt}, \omega, q}) \lambda_j^q \leq p^\omega, \quad \forall t, \omega \quad (\text{S8d})$$

$$\sum_{q \in Q_j} \lambda_j^q = 1, \quad \forall j \in S \quad [\sigma_j] \quad (\text{S8e})$$

$$\lambda_j^q \geq 0, \quad x_0 \geq 0, \quad (\text{S8f})$$

the first three rows holding for every index of (S1b)–(S1d).

The constant $\bar{\gamma}$ of Proposition 2 is measured on the scenario-expanded loss. For $j \in N$ and $\bar{x}_j \in \text{conv}(\mathcal{X}_j^{\text{MIP}})$, let $\bar{a}_j^{\text{bal}} = (\bar{a}_{jt}^{k, \text{bal}, \omega})_{k, t, \omega}$ be its net community position in every carrier, period and scenario, let $\mathcal{X}_j(\bar{a}_j^{\text{bal}})$ be the schedules of j attaining that position in every (k, t, ω) , and write $\bar{w}_j$, $\bar{r}_{jt}^{u, \omega}$, $\bar{r}_{jt}^{d, \omega}$ and $\bar{n}_{jt}^{E, \omega}$ for the expected objective value, the reserve contributions and the net market imports of $\bar{x}_j$, and $w_j(x)$, $r_{jt}^{u, \omega}(x)$, $r_{jt}^{d, \omega}(x)$, $n_{jt}^{E, \omega}(x)$ for the same quantities at x . With $\bar{\Pi}^{\text{res}} := \sum_{i \in I} \bar{\pi}_i^{\text{res}}$, put

$$L_j^\Omega(x) := (\bar{w}_j - w_j(x))^+ + \bar{\Pi}^{\text{res}} \max_{t \in T, \omega \in \Omega} \{ (\bar{r}_{jt}^{u, \omega} - r_{jt}^{u, \omega}(x))^+, (\bar{r}_{jt}^{d, \omega} - r_{jt}^{d, \omega}(x))^+ \} + \delta^{\text{peak}} \sum_{\omega \in \Omega} \rho_\omega \max_{t \in T} (n_{jt}^{E, \omega}(x) - \bar{n}_{jt}^{E, \omega})^+, \quad (\text{S9})$$

and define $\tilde{\gamma}_j$ and $\bar{\gamma}$ by (27) with L_j^Ω in place of L_j and $\mathcal{X}_j(\bar{a}_j^{\text{bal}})$ as above. The reserve rows (S1c) are imposed in every scenario, so a shortfall anywhere in $T_i \times \Omega$ caps r_i^{sym} for the whole block, which is why that term is a maximum over t and ω and not a sum; the peak enters once per scenario and is charged in expectation, which is why the last term is.

Proposition S1. *For the model (S1), Assumption 2 holds, and*

- (a) *Lemma 1 applies to (S8), so the Lagrangian sub-game is a linear production game, and since $b_j = 0$ its Owen solution is σ_j^* . The allocation (23) of Γ^{MIP} is then $\chi_j^{\text{OW}} = \sigma_j^* - \text{val}(\text{DP}_{\{j\}}^\Omega)$, the stand-alone term being the scenario-expanded single-member problem;*
- (b) *Proposition 1 and Corollary 1 hold.*
- (c) *Proposition 2 holds with m given by (S5) and with $\bar{\gamma}$ measured on (S9), which is independent of $|\Omega|$ under the price support fixed in Section S2-A, so*

$$\varepsilon^{\text{LR}} \leq \frac{(m+1)\bar{\gamma}}{n} = O\left(\frac{|T||\Omega|}{n}\right). \quad (\text{S10})$$

Proof: Assumption 2 was verified after (S5), and (S2)–(S4) place the model in the form the proofs of the main paper use, so (a) and (b) follow with no change of argument. For (c), the construction of Proposition 2 is repeated with L_j^Ω , and only two points require comment.

First, $\mathcal{X}_j(\bar{a}_j^{\text{bal}})$ is nonempty. Take any $(x_j, y_j) \in \mathcal{X}_j^{\text{MIP}}$, keep its first-stage indicators and its physical variables, and

in each scenario set the community trades to $\bar{a}_j^{\text{bal}}$ and close (11e)–(11f) with the market trades, which (13) admits scenario by scenario. Neither the community nor the market trades enter the commitment rows (S1e)–(S1h), so feasibility is preserved.

Second, each term of (S9) is bounded independently of $|\Omega|$. The objective enters as the expectation (S3), whose value is bounded by the capacities of j and by $\max_{t, k, \omega} \pi_t^{k, \text{imkt}, \omega}$ because $\sum_{\omega} \rho_\omega = 1$ and the prices lie in a fixed compact set, so refining the scenario set enlarges neither. The reserve term is a maximum over t , over the two directions and over ω , and not a sum, hence bounded by the reserve capacity of j . The peak term is an expectation over ω of a maximum over t , bounded by the trade bounds (11h) for the same reason. Hence $\bar{\gamma}$ is finite and depends on the assets of j alone. The Shapley–Folkman argument is then applied in $\mathbb{R}^{m+1}$ with m as in (S5). ■

Remark S2 (The bound is one-sided in $|\Omega|$). The factor $|\Omega|$ in (S10) enters through m , the number of linking rows, and bounds the number of members that Shapley–Folkman leaves at a fractional schedule. It should not be read as a statement about ω^{LR} itself. A finite Ω is a discretization of an underlying distribution, and refining it makes (DP_S^Ω) and (DWR_S^Ω) approach the problems it approximates, so the gap between them has no reason to grow with the number of scenarios while the bound does. What (S10) supports is the same structural statement as in the deterministic case, that the count of fractional members is set by the linking rows and not by n , and the decay in n that follows from it.

Remark S3 (Cost of the extension). The pricing problem (S7) remains one mixed-integer program per prosumer, of $|\Omega|$ times the size of its deterministic counterpart, and the n of them stay independent. The separation (30) of the row generation grows by the same factor while remaining a single monolithic program, so the comparison of Table I is unaffected.